Stable and Verifiable Latent Dynamics for World Models: A Sampling-to-Proof Gap Audit and an Empirical Failure Atlas of Stabilization Attempts

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***Abstract*—World models of the Dreamer family learn latent dynamics with no stability constraint on the imagined trajectory: long-horizon imagination can drift, and formal certification of the latent transition map is complicated by the categorical latent output. This paper makes three contributions toward stable and verifiable latent dynamics. First, we introduce a sampling-to-proof gap audit for the one-step latent map $z \mapsto z'$ under a frozen policy: a Lyapunov decrease condition, a large random-sampling baseline, and a branch-and-bound verifier with a reachability gate that discards off-distribution counterexamples. Second, we measure a structural verifier wall: interval bound propagation fails on the categorical output (softmax / sum-division) at any useful box width, so every branch-and-bound box is recorded as *unknown* with its reason – never as a violation and never as certification. Third, we report an empirical failure atlas from a controlled latent-space study: five documented mechanisms by which the optimizer routes around scalar stability constraints in a self-supervised latent space, including the decoupling of a spectrum-sum constraint from the leading Lyapunov exponent and the decoupling of finite-time spectral proxies from the asymptotic exponent. From the atlas we derive design rules for stabilization attempts and an evaluation standard that gates attractor-overlap metrics on rollout aliveness. All claims are stated with the exact object, region, and condition that produced them; nothing is carried over across systems.**

I. Introduction

Model-based reinforcement learning via world models has become a general approach to control: DreamerV3 [?], [?] masters over 150 tasks from a single configuration, and its latent imagination is the mechanism by which the policy is optimized. The world model learns a Recurrent State-Space Model (RSSM) [?] whose latent state evolves under the imagined actions, and the policy is trained on trajectories imagined in that latent space, never in the environment. This is precisely why the *stability* of the latent dynamics matters: if the imagined trajectory drifts off-distribution, or diverges, or converges to a spurious fixed point, the policy is optimized against a model of a world that does not exist. DreamerV3 is trained without any constraint that the latent map be contractive, energy-bounded, or in any other sense stable, and the categorical latent output makes direct formal verification non-trivial.

Two research lines have begun to address this gap. The first constrains the *learned dynamics* so that imagination stays well behaved; the most recent example is Koopman Dreamer [?], which replaces the deterministic latent core with spectrally constrained rotation–scaling blocks of bounded radius and derives a multi-step rollout-error bound. The second verifies a *frozen* model after the fact using neural Lyapunov methods [?], [?] with sound bound propagation [?], [?]. Both lines face a common, largely unexamined problem: how large is the gap between what random sampling finds and what a verifier can prove, and can scalar or spectral constraints actually shape the stability of a learned latent map?

This paper addresses that problem. Its contributions are:

- **A sampling-to-proof gap audit for latent dynamics.** We certify the one-step latent map $z \mapsto z' = (d', s')$ of a frozen DreamerV3-size model under a deterministic actor, where s' is the *mean* of the prior categorical (the softmax output). The audit compares a large random-sampling baseline against branch-and-bound verification of a Lyapunov decrease condition, and gates any candidate violation on whether the model's own dynamics reach it (a reachability gate). Off-distribution counterexamples are discarded, not reported with a caveat.
- **A measured verifier wall on categorical latents.** Interval bound propagation cannot bind the categorical output at any useful width: the softmax / sum-division node asserts on non-positive relaxed denominators (measured: "AssertionError in BoundReciprocal" on the categorical mean). The audit probes this once and records every branch-and-bound box as *unknown* with the reason, so the wall is measured rather than assumed.
- **An empirical failure atlas of stabilization attempts.** In a controlled latent-space study (a Hamiltonian-JEPA predictor trained end to end on chaotic Lorenz-63 data under a nonlinear coordinate warp), we isolate five mechanisms by which the optimizer satisfies scalar stability constraints without producing stable dynamics: (1) trivial representation freeze, (2) scale-conditioning paradox, (3) coordinate-skew loophole plus a dead saddle in dissipation-matrix learning, (4) spectrum-sum versus leading exponent decoupling, and (5) finite-time spectral traps. Each is annotated at its mechanism site.

Manuscript draft, [TODO: month year]. This work reuses the DreamerV3 one-step latent map, the auto_LiRPA branch-and-bound verifier, and a Hamiltonian-JEPA latent-dynamics benchmark. All measured numbers reported here were produced on CPU with fixed seeds; the trained-model audit on a DeepMind Control Suite task is [TODO: status] and is required before any quantitative claim about DreamerV3 itself is made public.

- **Design rules and an evaluation standard.** From the atlas we derive concrete rules for stabilization attempts (what to constrain, what to initialize, what to gate) and argue that attractor-overlap metrics (Chamfer/Hausdorff distance to the encoded manifold) must be gated on rollout aliveness, because a frozen predictor posts near-perfect overlap while being dynamically dead.

Honest framing. This is a methods and measurement paper. The verifier wall and the failure atlas are measured results; the direct application to a *trained* DreamerV3 on a control task is [TODO: pending GPU audit], and no quantitative claim about DreamerV3 itself is made until that audit lands. The latent-space study uses Lorenz-63, which is not a natural GENERIC system; the energy, entropy, and dissipation quantities learned there are proxies, and we claim on-attractor stability and invariant recovery, not recovery of true thermodynamics.

II. Background and Problem Setting

A. The RSSM and the one-step latent map

DreamerV3's world model factorizes the transition into a deterministic recurrent core and a stochastic posterior over discrete classes:

$$d_t = f_\theta(d_{t-1}, s_{t-1}, a_{t-1}), \qquad s_t \sim q_\phi(s_t \mid d_t, o_t), \tag{1}$$

with a prior $s_t \sim p_\phi(s_t \mid d_t)$ used at imagination time. In this paper the audited object is the one-step latent map under a frozen deterministic actor π:

$$\boldsymbol{z} = (d, s) \;\mapsto\; \boldsymbol{z}' = \big(d',\ \mathrm{softmax}(\mathrm{logits}')\big), \qquad d' = \mathrm{core}\big(d, \pi(\boldsymbol{z})\big) \tag{2}$$

where the stochastic output is replaced by the *mean* of the prior categorical, and $\pi(\boldsymbol{z}) = \tanh(\mathrm{mean}(\mathrm{actor}))$. The model is audited as trained; only the Lyapunov candidate V moves. Nothing about the imagined rollout is certified, and nothing larger than the DreamerV3 size-1M configuration (deter 512, hidden 64, stoch 32×4 classes) is considered.

B. Why stability of this map matters

The policy is optimized on imagined trajectories driven by the prior transition. If the map in (**??**) admits expanding directions, long-horizon imagination accumulates error and the imagined distribution drifts from the posterior distribution seen at training time – the posterior/prior mismatch that recent work attempts to reduce [**?**], [**?**]. Stability of the latent map is therefore not a theoretical nicety but a prerequisite for reliable planning in real systems, and for any safety argument about policies trained on imagined data.

C. Verification machinery

We verify the Lyapunov decrease condition

$$\mathrm{cond}(\boldsymbol{z}) = V(\boldsymbol{z}) - V(\boldsymbol{z}') > 0 \tag{3}$$

over a box region $\mathcal{B}$ built from the model's own on-policy latent states. Sampling draws $\boldsymbol{z} \sim \mathcal{B}$ and evaluates (**??**) exactly. Branch-and-bound uses CROWN-style [**?**] interval and linear relaxation bounds as implemented in auto_LiRPA [**?**]; the verifier source is pinned by the SHA-256 of the driver (the exact artifact that produced a result). A verdict is one of three: *violation* (a counterexample the model demonstrably reaches), *certified* (the bound proves the condition on the whole box), or *unknown* (bounds stayed loose, no counterexample found). *unknown* is verifier incompleteness: it is never evidence of safety, and it is never a gap and never a partial finding.

D. Related work on latent-space structure

On the representation-learning side, JEPA-style objectives [**?**], [**?**] use anti-collapse regularizers such as SIGReg (sketched isotropic Gaussian regularization via characteristic-function matching) to keep latent embeddings full-rank without a teacher network. The latent-space study in Section **??** builds directly on this line: it asks what happens when one additionally imposes *physical* structure (Hamiltonian energy conservation, metriplectic dissipation) on the latent dynamics of such a model. On the dynamics side, spectral constraints on recurrent dynamics [**?**] and the recent Koopman Dreamer backbone [**?**] bound modal radii to control error accumulation; Section **??** connects the failure atlas to exactly these designs.

III. The Sampling-to-Proof Gap Audit

The audit answers one question: are there latent states where a large random-sampling audit of the decrease condition (**??**) finds nothing, yet branch-and-bound finds a genuine violation that the model's own dynamics actually reach? If yes, that state is a sampling-to-proof gap and the strongest possible evidence about the limit of empirical evaluation.

A. Region construction

The region is built *from* the model's own latent states: a frozen checkpoint is rolled out on-policy and the posterior latents are recorded; the certified region is the coordinate box around those latents. The region is therefore on-distribution by construction. The certified map (**??**) is the prior-mean surrogate; this distinction is recorded in the artifact and never conflated.

B. Procedure

The branch-and-bound region is an annulus around the latent fixed point $\boldsymbol{z}^*$, decomposed into $2k$ boxes by holing the k coordinate dimensions with the largest halfwidths, so the number of boxes does not scale with the latent dimension. The annulus excludes the fixed point itself, where $\mathrm{cond}(\boldsymbol{z}^*) = 0$ by construction.

IV. The Verifier Wall on Categorical Latents

The certified graph contains the categorical output $s' = \mathrm{softmax}(\mathrm{logits}')$, i.e. an $\exp / \sum \exp$ with a division by the relaxed sum. Interval bound propagation on this node asserts whenever the relaxed logits interval spans a region where the softmax denominator can be non-positive in the bound computation. At the measured vacuity of the intermediate bounds, this happens on the first box ("AssertionError in BoundReciprocal"), and the *same* structural failure repeats

Algorithm 1 Sampling-to-proof gap audit

Require: Frozen model, on-policy latent support $\mathcal{S}$, box $\mathcal{B} \supseteq \mathcal{S}$, Lyapunov candidate V

1: Fit V to minimize (**??**) violations on samples from $\mathcal{B}$ (gradient training; the model is frozen)
2: **Sampling baseline:** draw N i.i.d. samples from $\mathcal{B}$, evaluate (**??**); record worst violation and count
3: **Reachability gate:** for each candidate violation, verify the model's closed-loop dynamics reach it (trajectory through the region to the candidate)
4: **Branch-and-bound:** decompose $\mathcal{B}$ into annulus boxes around the latent fixed point; run the verifier per box with a budget
5: **Wall gate:** probe the categorical output once; if the interval pass asserts on softmax/sum-division, record every box as *unknown* with the reason
6: **Verdict:** `gap_demonstrated` = sampling found nothing *and* branch-and-bound found a reachable violation

identically on every box. Because the bound-graph build dominates the per-box cost (measured: ~4 minutes for the graph trace on a size-1M graph on CPU), running the full branch-and-bound without the wall gate wastes hours to rediscover the same assertion.

The audit therefore probes the wall once, then records every box as *unknown* with the reason "categorical output not interval-bindable at this width". This is the guardrail-predicted boundary of the verification toolchain, now *measured* rather than assumed. The sampling baseline is the empirical result; a sampling violation is the only thing that can become a finding, and it must survive the reachability gate first.

On the smoke configuration (weight-initialized surrogate, not a trained model), the full pipeline runs end to end on CPU: the latent fixed point converges (residual 2.98×10^{-8}), the region holds 83.8% of steady-state latents, sampling finds 0/12 866 violations (worst condition $+2.2\times10^{-3}$), and the wall is recorded with `gap_demonstrated` = `false`. *None of these numbers describe a trained model*; they establish that the pipeline is sound and reproducible, which is their only claim. [TODO: trained-model audit on dm_control cartpole_balance; expected verdict on the categorical wall is *unknown*, which is verifier incompleteness and not a finding.]

V. The Failure Atlas of Latent-Space Stabilization

The wall says the frozen model cannot be certified directly. The remaining question is whether it can be *stabilized* – and here we report the empirical atlas from a controlled study: a Hamiltonian-JEPA predictor trained end to end from data on chaotic Lorenz-63 under a nonlinear coordinate warp, in five ablation arms (B/C/D/E/F) with matched data and steps. The study is fully reproducible on CPU (fixed seeds, one command). The atlas is the *negative result*: imposing physical / thermodynamic constraints inside a self-supervised latent space does not, by itself, yield physical dynamics, because the optimizer routes around each scalar constraint through a geometric exploit.

TABLE I
Documented failure modes of latent-space stabilization (Lorenz-63, nonlinear warp; reproduced on CPU in progress at draft time).

#	Mode	Measured signature
1	Representation freeze	motion $\sim 10^{-3}$ vs target ~ 0.4; λ_1-penal
2	Scale-conditioning paradox	raw proxy spread ~ 0.036 vs $\|\partial H/\partial p\|$
3	Skew loophole + dead R saddle	$\mathrm{tr}(R)/n = 0.00000$ in every naive run; R
4	Sum-vs-λ_1 decoupling	contraction -12.8 while λ_1 blows up to
5	Finite-time spectral trap	short-horizon $\hat{\lambda}_1 \to -0.17$ while true 200

Each mode is annotated at its mechanism site in the study source; the headlines are:

(1) Freeze. Multi-step alignment is minimized by a predictor that does not move: compounding K-step error makes motion locally costlier than none, and the field gradients collapse. A horizon curriculum and a detached motion-magnitude penalty are partial mitigations.

(2) Scale-conditioning paradox. Matching the raw magnitude of a kinetic energy proxy clamps the dynamics: the proxy's spread is $\sim 30\times$ below what the integrator needs, so magnitude matching freezes the model. Comparing *z-scores* of both sides – keeping only the shape/correlation constraint – restores $O(1)$ loss and frees the scale.

(3) Skew loophole and dead R saddle. A state-dependent skew $\mathbf{J}(\boldsymbol{z})$ is not divergence-free, so it can supply demanded volume contraction with no dissipation at all, leaving $\mathbf{R}$ off; and $\mathbf{R} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ zero-initialized has $\partial\mathbf{R}/\partial\mathbf{L} = 0$ at $\mathbf{L} = 0$, a dead saddle no gradient can lift. Both must be closed (fixed constant canonical skew; small nonzero $\mathbf{L}$ init) before dissipation engages ($\mathrm{tr}(R)/n \sim 5.6$, contraction -13.667).

(4) Sum-vs-λ_1 decoupling. Pinning the divergence pins the *sum* of the Lyapunov spectrum – an invariant – but the sum does not pin any individual exponent. Model E reaches strong contraction while λ_1 blows up to $+3.5$ (true ~ 0.88): the negative exponents compensate. Confirmed four times ($\lambda_1 \in \{-0.42, +1.44, +3.5, +4.29\}$, all with the "correct" sum). A scalar volume constraint is structurally incapable of shaping the leading exponent. This is the direct caution for spectral designs that constrain aggregate spectrum information.

(5) Finite-time spectral trap. At short horizons the finite-time QR exponent is decoupled from the asymptotic λ_1: training drives $\hat{\lambda}_1$ to -0.17 while the true 200-step exponent is $+1.44$ (opposite sign). At long horizons the proxy is trainable but nothing constrains global boundedness: the eval rollout diverges to $+\infty$. The trap moves from "wrong proxy" to "right proxy, unbounded global trajectory".

The overlap trap (evaluation standard). Chamfer/Hausdorff distance to the encoded attractor *rewards* a frozen predictor: a stationary point inside the attractor cloud scores near-perfect while being dynamically dead. Geometric overlap must be gated on the rollout being alive and chaotic ($\lambda_1 > 0$ and a sane motion ratio), or replaced by a motion-invariant statistic.

Transfer caveat. The atlas is measured on Lorenz-63 latent dynamics, not on DreamerV3's RSSM latents. Its role here is to make the failure modes *concrete and checkable*; the

D1 audit (Section **??**) is the tool that tests whether the same exploits appear in a trained world model's latent space. That test is the paper's primary open item.

VI. Design Rules for Stabilizing Latent Dynamics

From the wall and the atlas we derive the following rules. Each is stated with the failure mode it is designed to avoid.

1) **Certify a verifiable surrogate, not the categorical output.** The categorical output is not interval-bindable at useful width (Section **??**). Any certification must target the deterministic component of the latent map, or a structurally verifiable surrogate, with the stochastic output treated as the documented boundary of the toolchain.
2) **Constrain modes, not sums.** A scalar volume/spectrum-sum constraint cannot shape the leading Lyapunov exponent (failure 4). Spectral designs should constrain per-mode radii – as in Koopman Dreamer [**?**] – and must verify the *leading* exponent directly, not the aggregate.
3) **Break dead saddles with principled initialization.** Any factorized dissipation or learned structure that starts at zero and is penalized toward engagement needs a nonzero initialization and a gradient-path check (failure 3).
4) **Use shape constraints, not raw magnitudes.** Scale-free (z-score / correlation) anchors decouple magnitude from shape and prevent scale-starvation (failure 2).
5) **Curriculize the horizon and gate evaluation on aliveness.** Short-horizon objectives freeze the predictor (failure 1) and finite-time proxies decouple from asymptotic exponents (failure 5); evaluation must include a long-horizon rollout, an aliveness gate, and a leading-exponent check, and must never report attractor overlap without it (overlap trap).
6) **Report the gap honestly.** *unknown* verdicts, loose bounds, and walls are verifier incompleteness. They are never evidence of safety and never a gap. Only a reachable violation counts, and it must survive a reachability gate.

VII. Discussion and Limitations

What is measured. The verifier wall is a measured property of auto_LiRPA 0.7.2 on the categorical output, reproducible on CPU in minutes. The failure atlas is a measured, reproducible negative result on a controlled latent-space benchmark. The audit pipeline is measured end to end on a surrogate.

What is not yet measured. A trained DreamerV3 model's latent dynamics have not yet been audited; that audit requires a GPU training run and is the paper's primary open item. The transfer of the failure modes from Lorenz latents to RSSM latents is a hypothesis the audit is designed to test, not a claim. *Unknown* verdicts on the categorical wall are expected and are not findings.

Scope. The certified object is the one-step prior-mean map under a frozen policy. Nothing about imagined multi-step rollouts is certified, and nothing about the stochastic posterior. The region is built from on-policy posterior latents; the certified map is the prior-mean surrogate. These distinctions are recorded in every artifact and never conflated.

Relationship to the Dreamer team's problem. World-model-based control deployed on physical systems needs latent trajectories that stay bounded and on-distribution over planning horizons, and needs a way to argue about that formally. This paper contributes the measurement machinery (the gap audit), the obstacle (the categorical wall), and the cautionary evidence (the failure atlas) that any stabilization design must confront. It does not claim to know the team's internal priorities; it claims only that these are the questions any stability argument for a Dreamer-style latent space must answer.

VIII. Conclusion

We presented a sampling-to-proof gap audit for the latent dynamics of Dreamer-style world models, measured the structural wall that interval bound propagation hits on categorical latent outputs, and reported an empirical failure atlas of stabilization attempts in latent space together with design rules and an evaluation standard derived from it. The contribution is honest measurement and a checkable negative result, in a form that makes the obstacles to stable, verifiable latent dynamics concrete. The next step – the trained-model audit – is a GPU run, and the atlas provides the exact hypotheses it should test.

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Draft status. Authors TODO; references to verify; trained-model audit [TODO: GPU run]; atlas reproduction on CPU in progress at draft time. Nothing in this draft has been submitted or peer reviewed.